

Python 3: Getting Started

Data and File Structures Laboratory

<http://www.isical.ac.in/~dfslab/2019/index.html>

Running Python

```
print("Hello world")
```

- Interactively via the `python3` or `ipython` interpreter, **OR**
- Write your program in a file
 - start your file with

```
#!/usr/bin/env python3
```
 - **or** run your file as

```
$ python3 <your-file-name>
```

Base types

- `int` 783 0 -192 0b010 (binary) 0o642 (octal) 0xF3
- `float` as for C
- `bool` True False
- `str` sequence of *characters* (Unicode allowed)
 - delimiters: `"..."` also `'...'` `"""..."""`
 - multiline strings allowed with `"""`, e.g.

```
"""String  
across  
lines"""
```
- `bytes` sequence of byte values (0-255) `b"\xfe\375abcd"`

Differences with C highlighted

Operators

- `and, or, not` instead of `&&, ||, !`
- `/` is primary-school division, e.g., $1/2 \rightarrow 0.5$
- `//` is C-like (integer) division, e.g., $1//2 \rightarrow 0$
- `**` is exponentiation, e.g., $2**4 \rightarrow 16$
- No increment `++` / decrement `--` operators
- others as for C (mostly)

Differences with C highlighted

Container types

- Tuples (11, "y", 7.4)
- Lists [11, "y", 7.4]
- str, bytes
- Dictionaries { 1:11, 2:"y", 3:7.4, 3.14:"pi" }
 - elements accessed using *keys*
- Sets { 11, "y", 7.4 }

elements accessed using indices

Sequences (tuples, lists, strings, bytes)

Indexing

negative indexes

1 =

positive indexes

gablu **Slicing**

-n	-(n+1)	...	-2	-1
[10,	20,	...	40,	50,]
0	1	...	n-2	n-1

- `range(n)`
- `range(start, stop, step)`
- `enumerate(l)`

Assignments

```
a = b = c = 0
```

```
y, z, r = 9.2, -7.6, 0
```

 multiple assignments

```
a, b = b, a
```

 convenient swap

```
+=      -=      *=      /=      %=
```


- `if <condition> :, elif <condition> :, else :`
- `while <condition> :, continue, break`
- `for <variable> in <container type> :`
`for <variable> in <iterator> :`
- `def fname(arg1, arg2, arg3) :`
(remember to include a `return <value>` or `return None`)

Scope is defined by spaces not braces!

Configure your editor appropriately.

■ Containers:

Membership `in`

`len()`, `max()`, `min()`, `sum()`, `sorted()`, `reversed()`

■ **Ordered containers** (lists, tuples, strings, bytes):

`l.index(val)`, `l.count(val)`

+ concatenation

■ **Lists:**

`append(value)`, `extend(sequence)`, `insert(index, value)`,
`sort()`, `reverse()`

- Read using `s = input("Prompt string: ")`
- `s` is always a string
- Convert `s` to appropriate type using `int()`, `float()`, `bool()`, etc.

Mutable vs. immutable types

- **Immutable:** str, bytes, tuples, frozensets
- **Mutable:** lists, dictionaries, sets

- <https://perso.limsi.fr/pointal/python:memento>
- <https://perso.limsi.fr/pointal/python:abrege>
- <http://ehmatthes.github.io/pcc/cheatsheets/README.html>